

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I N.JOE JERISH (192524374) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle (Ambulance), EmergencyType (Ambulance), Vehicle (CarA), Behind (CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench } [1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$
- 2) $\text{Breeze } [1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$
- 3) $\text{Glitter } [x, y] \rightarrow \text{Gold is at } [x, y]$
- 4) $\neg \text{Wumpus } [x, y] \wedge \neg \text{Pit } [x, y] \rightarrow \text{Safe } [x, y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — Smart Medical Diagnosis System

Part (a)

Symbol definitions:

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Fl	Possible Flu
M	Possible Measles
P	Possible Pneumonia

Rules in Propositional Logic:

Rule i: $F \wedge C \rightarrow Fl$

Rule ii: $F \wedge R \rightarrow M$

Rule iii: $C \wedge B \rightarrow P$

Facts for Patient A:

F = True, C = True, B = True (R is unknown)

Part (b)

Forward Chaining — deriving all possible diagnoses for Patient A:

Step	Rule Fired	Condition Check	Fact Derived
1	—	Initial facts asserted	F = True, C = True, B = True
2	Rule i: $F \wedge C \rightarrow Fl$	F=True and C=True ✓	Fl (Possible Flu) derived
3	Rule ii: $F \wedge R \rightarrow M$	F=True but R=Unknown ✗	Rule cannot fire — M NOT derived
4	Rule iii: $C \wedge B \rightarrow P$	C=True and B=True ✓	P (Possible Pneumonia) derived

Final derived diagnoses for Patient A: Possible Flu (Fl) and Possible Pneumonia (P).

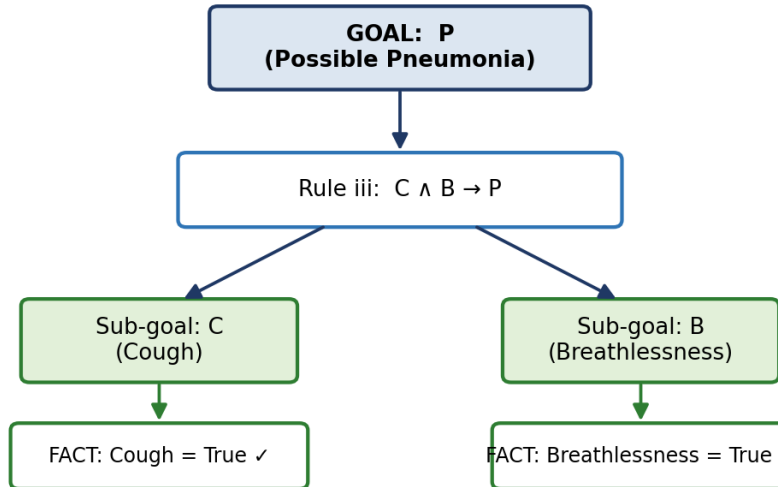
Part (c)

Backward Chaining — goal: 'Patient A has Pneumonia' (Goal = P):

- Start at goal P. Find a rule whose conclusion is $P \rightarrow$ Rule iii: $C \wedge B \rightarrow P$.
- Reduce goal P into two sub-goals: C (Cough) and B (Breathlessness).
- Check sub-goal C against the facts: Cough = True $\rightarrow$ sub-goal proved.
- Check sub-goal B against the facts: Breathlessness = True $\rightarrow$ sub-goal proved.
- Both sub-goals are satisfied, so the rule fires and the original goal P is proved TRUE.

Backward Chaining - Goal Reduction Tree
Goal: Patient A has Pneumonia (P)

Both sub-goals proven TRUE $\Rightarrow$ Goal P is CONFIRMED



CASE STUDY 2 — Autonomous Traffic Management Agent

Part (a)

Unification of Rule 1 with the given facts:

Rule 1: $\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$

Step	Unification performed	Substitution θ (MGU)
1	Unify $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$	$\theta_1 = \{x / \text{Ambulance}\}$
2	Apply θ_1 to $\text{EmergencyType}(x) \rightarrow \text{EmergencyType}(\text{Ambulance})$; matches fact directly	$\theta_2 = \{x / \text{Ambulance}\}$ (confirmed)
3	Final MGU for Rule 1	$\theta = \{x / \text{Ambulance}\}$

Result: Substituting θ into the rule conclusion gives $\text{ClearPath}(\text{Ambulance})$ — the rule successfully unifies and fires.

Part (b)

CNF conversion of the rules:

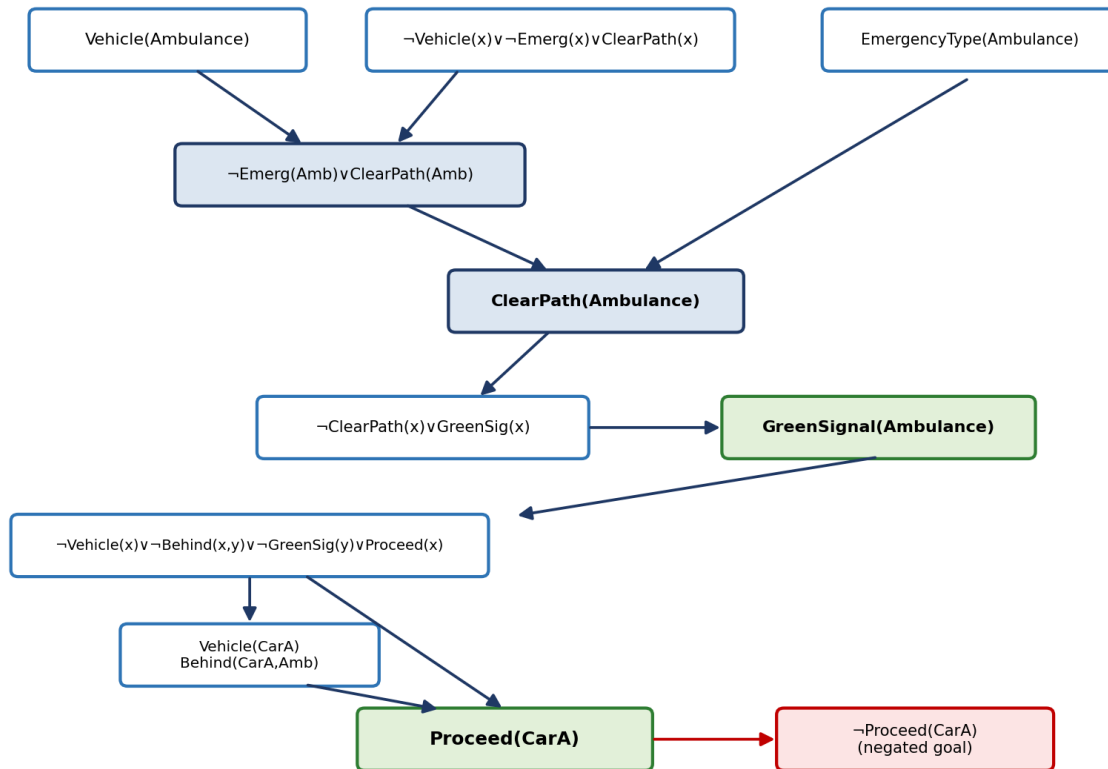
Rule	CNF Clause
Rule 1	$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
Rule 2	$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$ (and $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$)
Rule 3	$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$
Negated Goal	$\neg \text{Proceed}(\text{CarA})$

Resolution steps:

- Resolve Vehicle (Ambulance) with Rule 1 $(x/\text{Ambulance}) \rightarrow \neg\text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
- Resolve with fact EmergencyType (Ambulance) $\rightarrow \text{ClearPath}(\text{Ambulance})$
- Resolve ClearPath (Ambulance) with Rule 2 $(x/\text{Ambulance}) \rightarrow \text{GreenSignal}(\text{Ambulance})$
- Resolve Rule 3 $(x/\text{CarA}, y/\text{Ambulance})$ with Vehicle (CarA) $\rightarrow \neg\text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg\text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$
- Resolve with fact Behind (CarA, Ambulance) $\rightarrow \neg\text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$
- Resolve with GreenSignal (Ambulance) (derived above) $\rightarrow \text{Proceed}(\text{CarA})$
- Resolve Proceed (CarA) with the negated goal $\neg\text{Proceed}(\text{CarA}) \rightarrow \text{Empty Clause } \square$

Since resolution produces the empty clause, the negated goal leads to a contradiction $\Rightarrow$ 'Proceed(CarA)' is **PROVED TRUE**.

Resolution Proof Tree - Prove Proceed(CarA)



Resolves to Empty Clause $\square \rightarrow$ Proceed(CarA) is PROVED

Part (c)

Is the current FOL knowledge base sufficient for two simultaneous emergency vehicles?

Analysis: No — the KB is NOT sufficient. Rule 1 grants **ClearPath(x)** to any vehicle satisfying **Vehicle(x) ∧ EmergencyType(x)** independently. If two ambulances (or an ambulance and a fire truck) approach the same intersection simultaneously, both would independently satisfy Rule 1, and Rule 2 would then assert **GreenSignal(x)** for both — including for conflicting/crossing directions. The KB has no predicate for relative priority, arrival time, or route conflict, so it cannot decide which vehicle should actually get the green signal first.

Additional predicates/rules needed:

- Priority (x, p) — priority level p of emergency vehicle x (e.g., ambulance > fire truck > police escort).
- Conflict (x, y) — true when the paths of x and y intersect.
- ArrivalTime (x, t) or Distance (x, d) — to break ties between same-priority vehicles.
- New rule: $\forall x \forall y \text{ EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{Conflict}(x, y) \wedge \text{HigherPriority}(x, y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$

Justification: without such a conflict-resolution rule, the current KB is logically monotonic and would derive ClearPath for every qualifying vehicle, producing an inconsistent traffic state (two green signals on crossing roads) — a genuine gap in the knowledge base.

CASE STUDY 3 — Agricultural Expert Reasoning System

Part (a)

CNF conversion — eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute:

Sentence	Step 1: Eliminate $\rightarrow$ ($A \rightarrow B \equiv \neg A \vee B$)	Final CNF
P1: SoilDry $\rightarrow$ IrrigationNeeded	$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$	$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod	$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$	$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$	$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$	$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
Facts: SoilDry, CropWheat	already atomic	SoilDry; CropWheat (unit clauses)

(De Morgan's law $\neg(A \wedge B) \equiv \neg A \vee \neg B$ was applied while moving negation inward for P2 and P3; distribution was not further needed since each clause is already a single disjunction.)

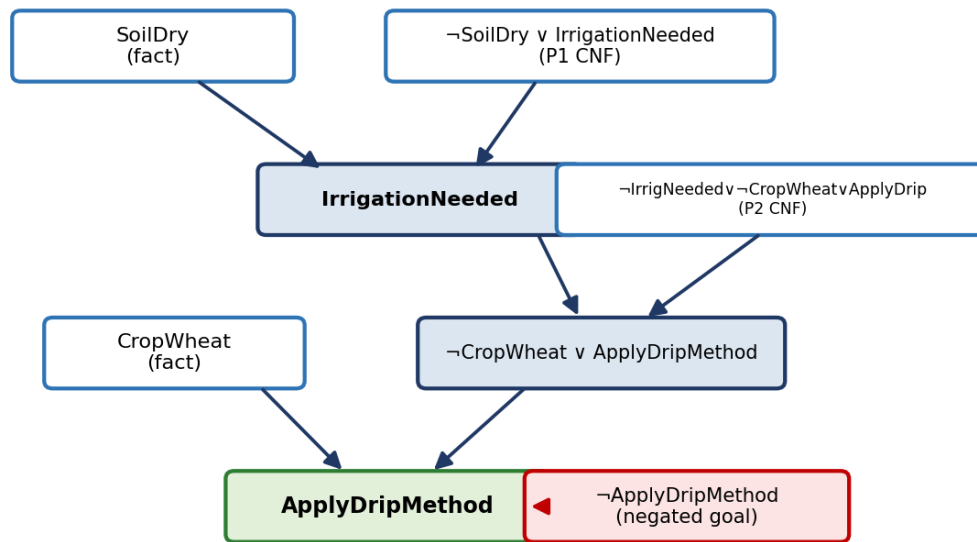
Part (b)

Resolution Refutation — prove 'ApplyDripMethod':

- Negate the goal: $\neg \text{ApplyDripMethod}$, and add it to the CNF knowledge base.
- Resolve SoilDry with $(\neg \text{SoilDry} \vee \text{IrrigationNeeded}) \rightarrow \text{IrrigationNeeded}$
- Resolve IrrigationNeeded with $(\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}) \rightarrow \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
- Resolve with fact CropWheat $\rightarrow \text{ApplyDripMethod}$
- Resolve ApplyDripMethod with the negated goal $\neg \text{ApplyDripMethod} \rightarrow \text{Empty Clause } \square$

The derivation of the empty clause proves, by refutation, that ApplyDripMethod logically follows from the KB.

Resolution Refutation Proof - ApplyDripMethod



Resolves to Empty Clause $\square \Rightarrow$ ApplyDripMethod is PROVED TRUE

Part (c)

If the fact 'SoilDry' is removed, how do the conclusions change?

- Without SoilDry as a unit clause, the clause $(\neg \text{SoilDry} \vee \text{IrrigationNeeded})$ can no longer be resolved down to IrrigationNeeded — so IrrigationNeeded can no longer be proven.
- Consequently, ApplyDripMethod can no longer be derived from P2 either (its antecedent IrrigationNeeded is unproven).
- Under the Closed-World Assumption (CWA) — common in rule-based expert systems — anything that cannot be proven true is treated as false. So $\neg \text{ApplyDripMethod}$ is assumed True.
- Applying P3 $(\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk})$ with $\neg \text{ApplyDripMethod} = \text{True}$ and $\text{CropWheat} = \text{True}$ (Modus Ponens) $\rightarrow \text{CropAtRisk}$ is derived.

Conclusion: Yes — under the Closed-World Assumption, 'CropAtRisk' STILL follows (via negation-as-failure), even though 'ApplyDripMethod' no longer does. Under strict open-world FOL semantics, however, neither ApplyDripMethod nor CropAtRisk could be proven, since $\neg \text{ApplyDripMethod}$ itself is not formally entailed — the outcome is indeterminate without CWA.

CASE STUDY 4 — Wumpus World Knowledge Agent

Part (a)

Belief state from percepts — applying Modus Ponens:

Percept	Rule Applied (Modus Ponens)	Derived Conclusion
Stench [1,2]	$\text{Stench } [1,2] \rightarrow \text{Wumpus adjacent to } [1,2]$	Wumpus is in $\{[1,1], [1,3], [2,2]\}$
Breeze [1,1]	$\text{Breeze } [1,1] \rightarrow \text{Pit adjacent to } [1,1]$	Pit is in $\{[2,1], [1,2]\}$
Glitter [2,2]	$\text{Glitter } [x, y] \rightarrow \text{Gold at } [x, y]$	Gold [2,2] is TRUE

All derived conclusions: $WumpusAdjacent [1,2] \Rightarrow Wumpus \in \{[1,1], [1,3], [2,2]\}$; $PitAdjacent [1,1] \Rightarrow Pit \in \{[2,1], [1,2]\}$; $Gold [2,2] = True$.

Part (b)

Forward Chaining — narrowing down Wumpus and Pit locations:

- $[1,1]$ is the agent's starting square and is known safe by convention $\rightarrow$ eliminate $[1,1]$ from the Wumpus candidate set.
- Remaining Wumpus candidates: $\{[1,3], [2,2]\}$.
- $[1,2]$ has already been visited safely (the agent is alive there) $\rightarrow$ eliminate $[1,2]$ from the Pit candidate set.
- Remaining Pit candidate: $\{[2,1]\}$ $\rightarrow$ since it is the only candidate, Pit $[2,1]$ is confirmed.
- Apply Rule 4 ($\neg Wumpus [x, y] \wedge \neg Pit [x, y] \rightarrow Safe [x, y]$) to all cells not implicated by any Stench/Breeze inference, e.g. $[3,1]$, $[2,3]$, $[3,2]$, $[3,3]$ $\rightarrow$ marked Safe (Unknown but not threatened).

Wumpus World - Belief Map (rows/cols 1-3)

[1,3] Wumpus? (candidate)	[2,3] Unknown	[3,3] Unknown
[1,2] Visited / Safe Stench!	[2,2] Glitter $\rightarrow$ GOLD Wumpus? (candidate)	[3,2] Unknown
[1,1] Start / Safe Breeze!	[2,1] PIT! (inferred)	[3,1] Unknown

Part (c)

Is the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ safe to collect the gold?

Cell	Safety Evaluation	Verdict
[1,1]	Starting square, safe by convention	Safe ✓
[1,2]	Already visited without triggering Pit/Wumpus death	Safe ✓
[2,2]	Contains Gold (from Glitter); but still an unresolved Wumpus candidate — $[2,2]$ was not eliminated from the Wumpus set in part (b)	NOT confirmed safe ⚠

Conclusion: The path is safe for its first two steps, but the final step into [2,2] carries residual risk because the Wumpus has only been narrowed to {[1,3], [2,2]} and not fully eliminated from [2,2]. Using logical inference alone (Rule 4 needs $\neg \text{Wumpus}[x, y]$ AND $\neg \text{Pit}[x, y]$ to certify Safe), [2,2] cannot yet be certified Safe. The agent should gather one more percept (e.g., check for stench while adjacent to [1,3], or use the absence of stench at another neighboring cell) to eliminate the remaining ambiguity before proceeding — entering [2,2] now would be a calculated risk, not a logically guaranteed-safe move.
